

Este es el examen final del curso *Análisis y Diseño de Algoritmos*, 2025-2. El examen tiene 8 preguntas; otorga un total de 100 puntos. El examen es *individual* y no es permitido el uso de libros, apuntes ni equipos electrónicos.

Nombre y código: _____

	1.1	1.3	1.4	2.4	3.4	6.4	Total
1	/ 10						
2		/10					
3		/10					
4			/15				
5				/15			
6					/10		
7						/15	
8						/15	
Total	/10	/20	/15	/15	/10	/30	/100

Nota: En los numerales 2 y 6 se supone que los algoritmos involucrados son determinísticos (i.e., funciones totales computables).

1. (10 puntos) Para $n \in \mathbb{N}$, refute: $n^2 \in O(n \log n)$.
2. (10 puntos) Enuncie la propiedad de que la clase P es cerrada bajo concatenación y demuéstrela.
3. (10 puntos) Consider the following statements about the complexity classes P, NP, NP-hard, and NP-complete:
 - a. Every problem in P is also in NP.
 - b. An NP-hard problem must be solvable in polynomial time.
 - c. If an NP-complete problem is solved in polynomial time, then every problem in NP can also be solved in polynomial time.
 - d. An NP-complete problem is both in NP and NP-hard.

Which of the statements are true? Justify your selection.

4. (15 puntos) Diseñe un algoritmo recurrente que, dado un arreglo $A[0..N]$ de números enteros y un número entero T , determine los “subcolecciones” de A cuya suma es exactamente T . Determine su complejidad temporal y las suposiciones que hace para dicho cálculo.
5. (15 puntos) A *shuffle* of two strings X and Y is formed by interspersing the characters into a new string, keeping the characters of X and Y in the same order. For example, the string BANANAANANAS is a shuffle of the strings BANANA and ANANAS in several different ways.

Design a dynamic programming algorithm that solves the following problem:

Input: Arrays $A[0..M]$, $B[0..N]$, and $C[0..M+N]$, $M \geq 0$ and $N \geq 0$.

Output: Is C a shuffle of A and B ?

Explain the main features of the proposed design, including an analysis of its asymptotic behavior.

6. (10 puntos) A student claims to have proved that $P = NP$ using the following argument:

Since the VERTEXCOVER problem is NP-complete, and since we can always find a vertex cover by checking all subsets of vertices of size at most k of a total of n vertices, the problem is solvable in polynomial time. Therefore, VERTEXCOVER $\in P$ and because VERTEXCOVER is NP-complete, it follows that $P = NP$.

Your task is to disprove the student's claim by identifying the flaw in the argument. Choose the best explanation and justify your answer:

- The student incorrectly assumes that VERTEXCOVER is NP-complete; in fact, VERTEXCOVER is in P.
- The student incorrectly claims that checking all subsets of vertices of size at most k takes polynomial time; this exhaustive search runs in exponential time.
- The student wrongly asserts that solving one NP-complete problem in polynomial time implies $P \neq NP$.
- The student's argument fails because NP-complete reductions only go from problems in P to problems in NP.
- The student is mistaken because VERTEXCOVER cannot be reduced from any NP problem.

7. (15 puntos) Considere los siguientes lenguajes:

$$\begin{aligned} \text{SUBSETSUM} &= \{ \langle X = \{x_0, \dots, x_{N-1}\}, S \rangle \mid X \subseteq \mathbb{N} \wedge S \in \mathbb{N} \wedge (+x \mid x \in X : x) = S \} \\ \text{2PAR} &= \{ \langle Y = \{y_0, \dots, y_{M-1}\} \rangle \mid Y \subseteq \mathbb{N} \wedge (\exists A, B) A \subseteq Y \wedge B \subseteq Y \wedge A \cup B = Y \wedge A \cap B = \emptyset \\ &\quad \wedge (+a \mid a \in A : a) = (+b \mid b \in B : b) \} \end{aligned}$$

El lenguaje SUBSETSUM es el problema de decisión asociado a los conjuntos finitos de números naturales X que tienen un subconjunto cuya suma es exactamente un valor dado S . El lenguaje 2PAR es el problema de decisión asociado a los conjuntos finitos de números naturales Y que pueden ser particionados en exactamente dos subconjuntos A y B con la misma suma. Por ejemplo, $\langle \{1, 2, 4\}, 1 \rangle \in \text{SUBSETSUM}$ y $\langle \{1, 2, 4\}, 3 \rangle \in \text{SUBSETSUM}$, mientras que $\langle \{1, 2, 4\}, 8 \rangle \notin \text{SUBSETSUM}$. A su vez, $\langle \{1, 2, 3\} \rangle \in \text{2PAR}$ y $\langle \{1, 2, 3, 4\} \rangle \in \text{2PAR}$, mientras que $\langle \{1, 2, 4\} \rangle \notin \text{2PAR}$.

Demuestre: $\text{SUBSETSUM} \leq_P \text{2PAR}$.

8. (15 puntos) Con base en la definición de SUBSETSUM en el problema anterior, demuestre que SUBSETSUM $\in NP$. Justifique su respuesta.